

The AI-Enhanced Deliverable

How the Part 1 slide became this website — and exactly how AI was used

What this deliverable is

The hiring exercise asks for an AI-enhanced version of the Part 1 slide, in a format that can be embedded in a website. **This website is that deliverable.** The one-page slide (download) is the seed; the site enhances it into something a static slide can't be:

- **The data story** expands the slide's narrative into a full memo with an executive summary, findings, recommendations, and caveats — with a PDF version built from the same source.
- **The interactive map** turns the slide's hero map into an explorable Leaflet map: hover any hex for its capacity, toggle on the 3,419 underlying opportunity parcels, click one for its details.
- **Methods & Sources** documents every cleaning decision, assumption, and methodological choice — the audit trail a slide can never carry.
- **The whole site is embeddable and reproducible:** it renders with Quarto from a public repository, republishes automatically via GitHub Actions on every push, and every number regenerates from one command (`Rscript code/run_all.R`).

How AI was used

The exercise asks for AI-driven enhancements to the Part 1 deliverable, and that is where I used it. I designed and built the analysis — the capacity framework, the opportunity screen and its exclusion rules, the hot-spot methodology, and the pipeline behind them. **Claude (Anthropic)** was then used to enhance and harden the deliverable:

- **Enhancing the one-page slide into this website** — converting the slide's map, figures, and narrative into the Quarto site you're reading, including the interactive Leaflet version of the hero map.
- **Producing the slide layout itself** — the page composition that turns the pipeline's figures and numbers into the polished one-page PDF.
- **Auditing the code** — reviewing the pipeline for correctness and reproducibility, and checking the cartography programmatically, including colorblind-safety validation (deuteranopia, protanopia, tritanopia) of the palette before it touched a map.

What stayed human: the choice of story (plan-enabled capacity, not another choropleth), the framing against the Growth Policy rather than zoning, the analysis and its assumptions, the exclusion rules and their ethics, the recommendations, and the final read on every number before it shipped.

Why this transparency matters

A firm that publishes AI-polished work owes its clients clarity about where the analysis ends and the polish begins. Everything quantitative on this site traces to the reproducible pipeline in the repository; the AI acceleration is documented here; and the assumptions that drive the headline numbers are inspectable in one file (`code/00_setup.R`). That's the standard I'd bring to client work.